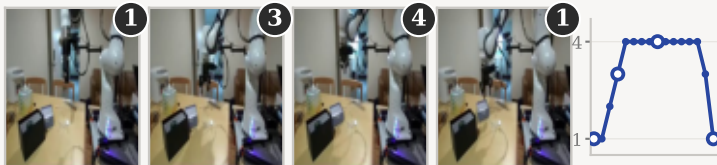


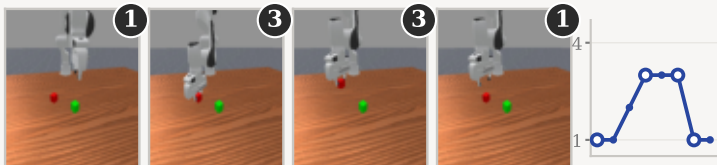
1 failure data, labeled densely

real-world · scored by a VLM+LLM pipeline



``Move lid on or off of container.'' (DROID)

simulated · generated and scored by design

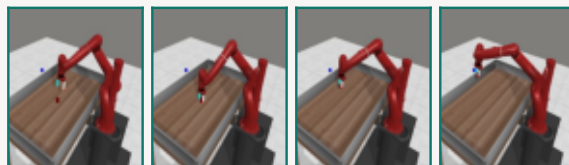


``Stack the red cube on the green cube.'' (FailSafe)

progress rises and collapses; the labels follow it

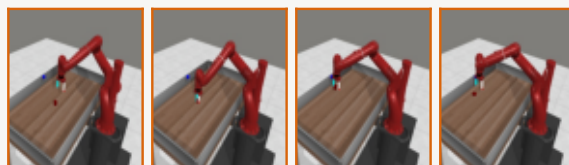
2 in-context demonstration

demonstration (success)



`<demo_end>`

query (same-task failure), loss here only



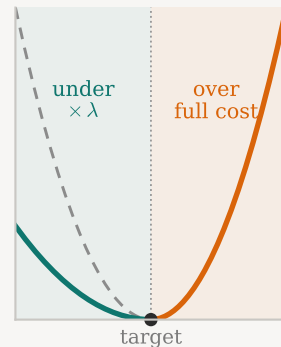
``Pick and place the puck.'' (MetaWorld)

3 asymmetric loss

Robometer-4B backbone



RoboRef



over-prediction is punished harder, so the reward is hard to exploit

4 zero-shot RL reward



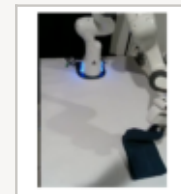
MetaWorld

in-distribution



LIBERO

Robometer's benchmark



Robomimic

out-of-distribution

policy trained on RoboRef's reward alone